## HW2 BPE tokenizer report - Diyar Husayyan, 214034621

**Implementation.** All three tokenizers use from-scratch character-up BPE with a mergeable space marker **S**, one-cross-space bigram cap, byte fallback that never emits UNK, pickle-safe state, and length-faithful decode for the fixed NER aligner. This differs from basic/pre-tokenized BPE by allowing selected cross-word merges while forbidding 3+ word tokens.

**Tokenizer choices.** tok1 targets Twitter: the balanced domain-1/domain-2 mix at vocab 6000 gave F1 0.4780, +0.043 over the earlier domain-1-only vocab-8000 tokenizer; larger mix vocabs fell sharply, and the structured pre-tokenization experiment was reverted after lowering F1. tok2 targets news with vocab 16000, keeping F1 0.9557 while giving the best measured efficiency/speed tradeoff among news candidates. tok3 handles the hidden domain with the balanced deduped mix, medium vocab 8000, and byte fallback; visible-domain checks are d1=0.4217, d2=0.9587, OOV stress 15/15, UNK=0.

| model/split | vocab | dev F1 | tok/char | tok/sent | cold tok/s |
|---|---|---|---|---|---|
| tok1 domain_1 | 6000 | 0.4780 | 0.307 | 22.52 | 24858 |
| tok2 domain_2 | 16000 | 0.9557 | 0.229 | 20.73 | 14821 |
| tok3 domain_1 | 8000 | 0.4217 | 0.294 | 21.55 | 28893 |
| tok3 domain_2 | 8000 | 0.9587 | 0.269 | 24.33 | 13750 |

**Top/bottom training bigrams.** Ordered BPE merge pairs from the shipped pickles; early pairs are the highest-count retained merges and late pairs are the lowest retained merges. **S** denotes the learned space marker.

| tok | top 5 merge pairs | bottom 5 merge pairs |
|---|---|---|
| tok1 | S+t; i+n; S+a; h+e; r+e | em+s; SB+ody; SO+N; Sg+ames; ,"+Sthe |
| tok2 | S+t; i+n; h+e; S+a; r+e | Sun+comfortable; Swith+,; Swith+. ; Swor+n; ,"+Sthe |
| tok3 | S+t; i+n; S+a; h+e; r+e | -+time; D+ay; z+a; S+ing; ,"+Sthe |

**Hidden-domain and model-2 discussion.** tok3 deliberately trades a little visible F1 for fewer tokens than vocab 4000 and broader mixed-domain coverage, with exact reconstruction and no UNK on stress cases. tok2's high news F1 reflects the cleaner news distribution and larger vocab; its 16000 setting improves compression/speed without materially sacrificing F1 versus smaller news tokenizers.